

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

A rectangular loop of sides 12 cm and 5 cm, with its sides parallel to the x-axis and y-axis respectively moves with a velocity of 5 cm/s in the positive x axis direction, in a space containing a variable magnetic field in the positive z direction. The field has a gradient of 10^{-3} T/cm along the negative x direction and it is decreasing with time at the rate of 10^{-3} T/s. If the resistance of the loop is $6\text{ m}\Omega$, the power dissipated by the loop as heat is $\text{_____} \times 10^{-9}$ W.

Q2 - 2024 (01 Feb Shift 2)

A coil of 200 turns and area 0.20 m^2 is rotated at half a revolution per second and is placed in uniform magnetic field of 0.01 T perpendicular to axis of rotation of the coil. The maximum voltage generated in the coil is $\frac{2\pi}{\beta}$ volt. The value of β is _____ .

Q3 - 2024 (27 Jan Shift 1)

A rectangular loop of length 2.5 m and width 2 m is placed at 60° to a magnetic field of 4 T. The loop is removed from the field in 10sec. The average emf induced in the loop during this time is

- (1) -2 V
- (2) $+2\text{ V}$
- (3) $+1\text{ V}$
- (4) -1 V

Q4 - 2024 (27 Jan Shift 1)

Two coils have mutual inductance 0.002H . The current changes in the first coil according to the relation $i = i_0 \sin \omega t$, where $i_0 = 5\text{ A}$ and $\omega = 50\pi\text{ rad/s}$. The maximum value of emf in the second coil is $\frac{\pi}{\alpha}\text{ V}$. The value of α is _____ .

Q5 - 2024 (29 Jan Shift 1)

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Questions with Answer Keys

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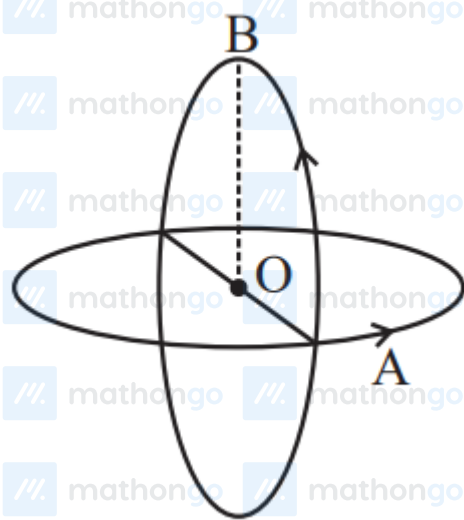
A square loop of side 10 cm and resistance 0.7Ω is placed vertically in east-west plane. A uniform magnetic field of 0.20 T is set up across the plane in north east direction. The magnetic field is decreased to zero in 1 s at a steady rate. Then, magnitude of induced emf is $\sqrt{x} \times 10^{-3}$ V. The value of x is _____

Q6 - 2024 (29 Jan Shift 2)

A horizontal straight wire 5 m long extending from east to west falling freely at right angle to horizontal component of earth's magnetic field $0.60 \times 10^{-4} \text{ Wbm}^{-2}$. The instantaneous value of emf induced in the wire when its velocity is 10 ms^{-1} is _____ $\times 10^{-3}$ V

Q7 - 2024 (30 Jan Shift 1)

Two insulated circular loop A and B radius ' a ' carrying a current of ' I ' in the anti clockwise direction as shown in figure. The magnitude of the magnetic induction at the centre will be :



(1) $\frac{\sqrt{2}\mu_0 I}{a}$

(2) $\frac{\mu_0 I}{2a}$

(3) $\frac{\mu_0 I}{\sqrt{2}a}$

(4) $\frac{2\mu_0 I}{a}$

Q8 - 2024 (31 Jan Shift 1)

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A small square loop of wire of side ℓ is placed inside a large square loop of wire of side L ($L = \ell^2$). The loops are coplanar and their centers coincide. The value of the mutual inductance of the system is $\sqrt{x} \times 10^{-7} \text{H}$, where $x =$ _____

Q9 - 2024 (31 Jan Shift 2)

The magnetic flux ϕ (in weber) linked with a closed circuit of resistance 8Ω varies with time (in seconds) as $\phi = 5t^2 - 36t + 1$. The induced current in the circuit at $t = 2 \text{ s}$ is _____ A.

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Questions with Answer Keys

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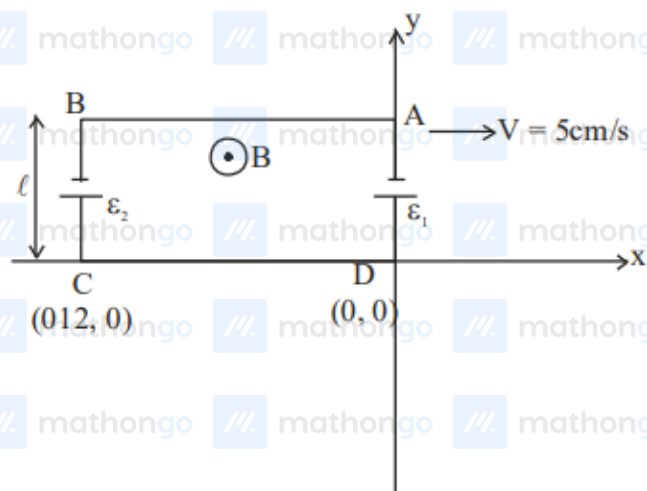
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Solutions

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Q1



B_0 is the magnetic field at origin

$$\frac{dB}{dx} = \frac{10^{-3}}{10^{-2}}$$

$$\int_{B_0}^B dB = - \int_0^x 10^{-1} dx$$

$$B - B_0 = -10^{-1}x$$

$$B = \left(B_0 - \frac{x}{10} \right)$$

Motional emf in AB = 0

Motional emf in CD = 0

Motional emf in AD = $\varepsilon_1 = B_0 \ell v$

Magnetic field on rod BC B

$$= \left(B_0 - \frac{(-12 \times 10^{-2})}{10} \right)$$

Motional emf in BC = $\varepsilon_2 = \left(B_0 + \frac{12 \times 10^{-2}}{10} \right) \ell \times v$

$$\varepsilon_{eq} = \varepsilon_2 - \varepsilon_1 = 300 \times 10^{-7} \text{ V}$$

For time variation

$$(\varepsilon_{eq})' = A \frac{dB}{dt} = 60 \times 10^{-7} \text{ V}$$

$$(\varepsilon_{eq})_{net} = \varepsilon_{eq} + (\varepsilon_{eq})' = 360 \times 10^{-7} \text{ V}$$

$$\text{Power} = \frac{(\varepsilon_{eq})_{net}^2}{R} = 216 \times 10^{-9} \text{ W}$$

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Solutions

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Q2

$$\phi = NAB \cos(\omega t)$$

$$\varepsilon = -\frac{d\phi}{dt} = NAB\omega \sin(\omega t)$$

$$\varepsilon_{\max} = NAB\omega$$

$$= 200 \times 0.2 \times 0.01 \times \pi$$

$$= \frac{4\pi}{10} = \frac{2\pi}{5} \text{ volt}$$

Q3

$$\text{Average emf} = \frac{\text{Change in flux}}{\text{Time}} = -\frac{\Delta\phi}{\Delta t}$$

$$= -\frac{0 - (4 \times (2.5 \times 2) \cos 60^\circ)}{10}$$

$$= +1 \text{ V}$$

Q4

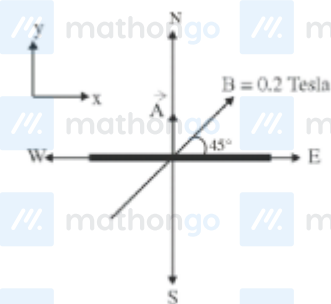
$$\phi = Mi = Mi_0 \sin \omega t$$

$$\text{EMF} = -M \frac{di}{dt} = -0.002 (i_0 \omega \cos \omega t)$$

$$\text{EMF}_{\max} = i_0 \omega (0.002) = (5)(50\pi)(0.002)$$

$$\text{EMF}_{\max} = \frac{\pi}{2} \text{ V}$$

Q5



$$\vec{A} = (0.1)^2 \hat{j}$$

$$\vec{B} = \frac{0.2}{\sqrt{2}} \hat{i} + \frac{0.2}{\sqrt{2}} \hat{j}$$

Magnitude of induced emf

$$c = \frac{\Delta\phi}{\Delta t} = \frac{\vec{B} \cdot \vec{A} - 0}{1} = \sqrt{2} \times 10^{-3} \text{ V}$$

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Solutions

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Q6

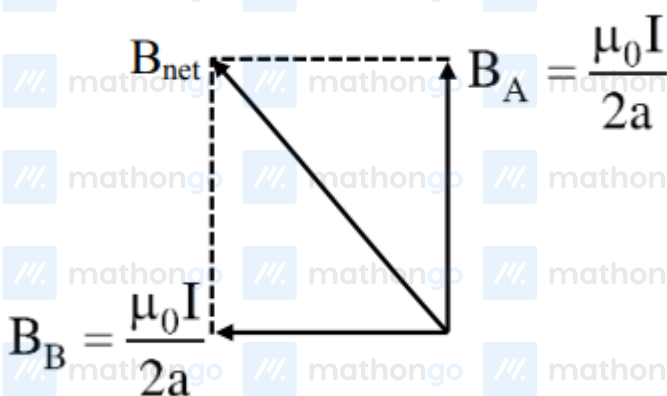
$$B_H = 0.60 \times 10^{-4} \text{ Wb/m}^2$$

$$\text{Induced emf } e = B_H v \ell$$

$$= 0.60 \times 10^{-4} \times 10 \times 5$$

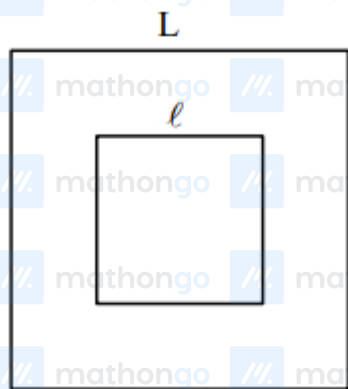
$$= 3 \times 10^{-3} \text{ V}$$

Q7



$$\therefore B_{net} = \frac{\sqrt{2}\mu_0 I}{2a}$$

Q8



Flux linkage for inner loop.

$$\phi = B_{\text{center}} \cdot \ell^2$$

$$= 4 \times \frac{\mu_0 i}{4\pi \frac{L}{2}} (\sin 45^\circ + \sin 45^\circ) \ell^2$$

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Solutions

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$$\phi = 2\sqrt{2} \frac{\mu_0 i}{\pi L} \ell^2$$

$$M = \frac{\phi}{i} = \frac{2\sqrt{2} \mu_0 \ell^2}{\pi L} = 2\sqrt{2} \frac{\mu_0}{\pi}$$

$$= 2\sqrt{2} \frac{4\pi}{\pi} \times 10^{-7}$$

$$= 8\sqrt{2} \times 10^{-7} \text{ H}$$

$$= \sqrt{128} \times 10^{-7} \text{ H}$$

$$x = 128$$

Q9

$$\varepsilon = - \left(\frac{d\phi}{dt} \right) = 10t - 36$$

$$\text{at } t = 2, \varepsilon = 16 \text{ V}$$

$$i = \frac{\varepsilon}{R} = \frac{16}{8} = 2 \text{ A}$$

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